Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3485**

Title: **EFFECTIVE SKIN WIDTH**

Revision: **1.0**

Date: **04/05/2020**

1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

This chapter defines the methodology to use for computing the effective skin width in composite stiffened panels subjected to compression load, when the skin is allowed to buckle between stringers (local buckling).

The effective width approach has traditionally supported the stress analysis of post-buckled stiffened metallic panels, subjected to compression loads, up to failure. This chapter permits the extension of this concept (see chapter §3185 for details regarding metal panels) to composite structures.

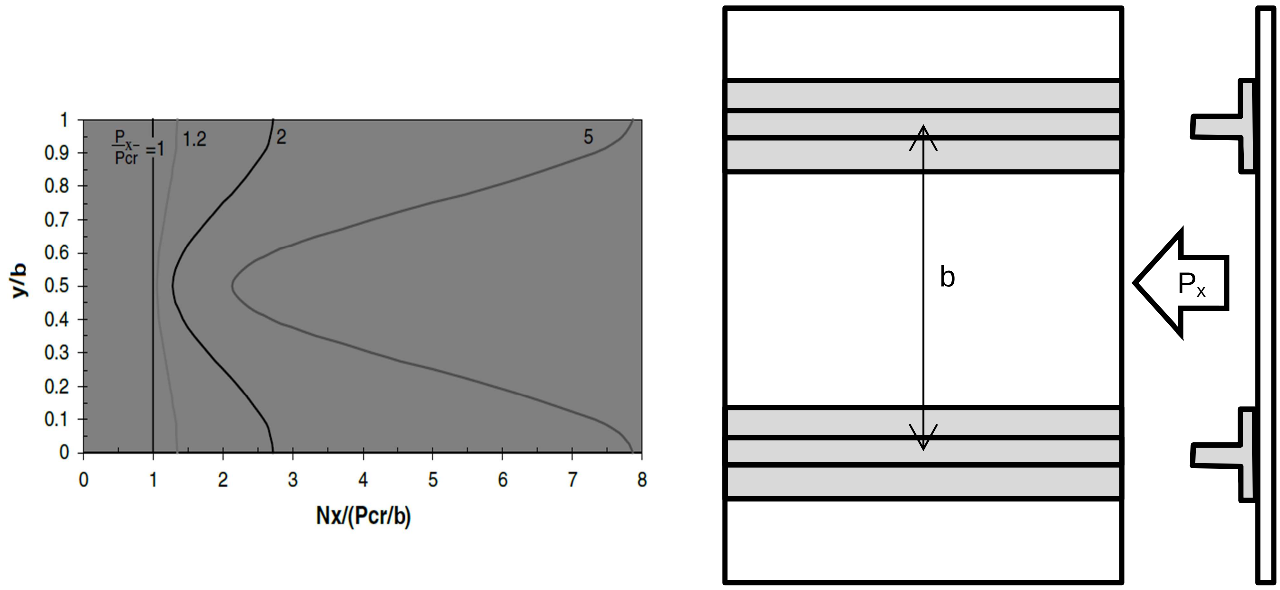

Figure 1. Compression stress distribution of post bucked skin

In tension, on longitudinal straight stringers, the whole skin width shall be considered effective for carrying axial loads. However, for the skin on curved frames, when subjected to hoop loads (eg. due to cabin pressurization), the calculation of the effective width is a more complex problem. As discussed by Schagerl (Ref. 3): *The skin of a curved aircraft panel flattens, if it is subjected to circumferential tension, and bulges, if it is subjected to compression. These deformations distort the hoop stress distribution, which further leads to a reduced load-carrying width of the skin in circumferential direction*.

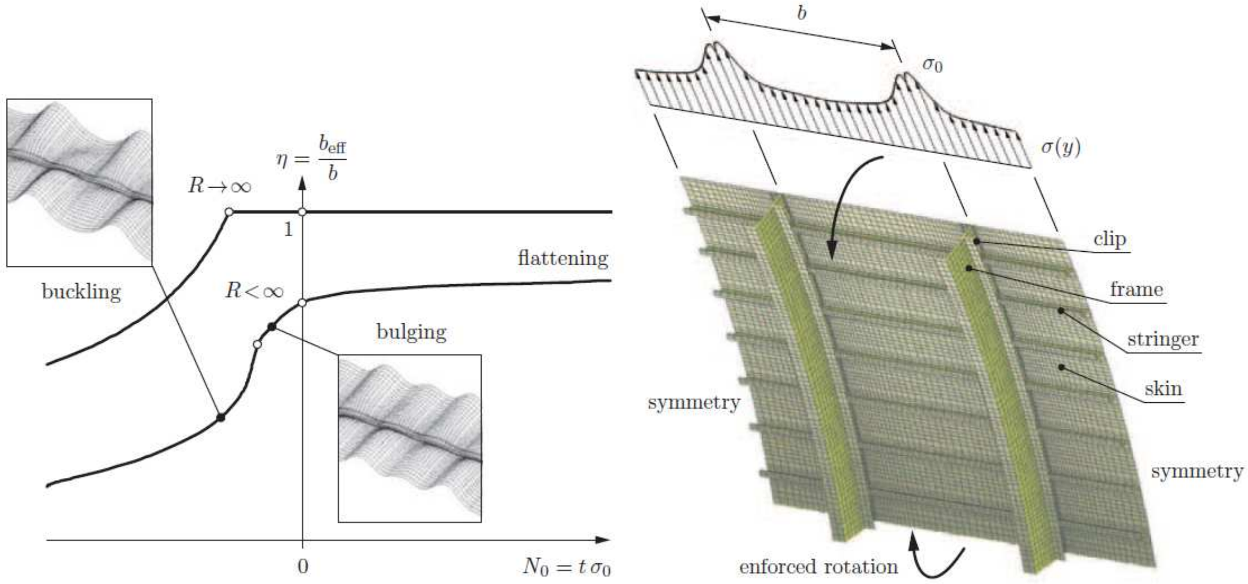

Figure 2. Flattening (tension) and bulging (compression) in curved skins at frame supports

Remark: The effective width problem of curved skins on frames shall not be treated in present revision of this chapter.

2 METHOD DESCRIPTION

2.1 EFFECTIVE WIDTH OF SKIN ON STRAIGHT STRINGERS

<u>*2.1.1 Legacy approach for effective width*</u>

Some Airbus legacy methods have taken advantage of the traditional Von Karman effective width approach, widely used in metals, to be applicable in composite stiffened skins in compression, by a straightforward translation of the familiar formulas to practical expression when working with composites (using the fundamental relationship between the buckling load and the panel width, i.e. $\varepsilon_{cr} \sim 1/w^2$, see discussion in Ref. 1), reaching the simple expression:

$$w_i = (\varepsilon_{cr}/\varepsilon_x)^{0.5} \cdot b / 2$$

Where:

w_i is the effective semi-width of skin (at each side of the stringer)

b is the real width of the skin (stringers pitch)

ε_{cr} is the compression strain at the buckling onset

ε_x is the compression strain in longitudinal direction

2.1.2 Effective width in homogeneous square panels

Some authors (e.g. Kassapoglou, see Ref. 2) have derived expressions to obtain this effective width as solution of the post-buckling Von Karman equations, in case of a square (flat) plate made of a composite orthotropic laminate (D_{16}=D_{26}=0), in compression:

$$w_i = \frac{b/2}{1 + 2\left(1 + \frac{A_{12}}{A_{11}}\right)\left(1 - \frac{P_{cr}}{P_x}\right)\left(\frac{A_{11}}{A_{11} + 3A_{22}}\right)}$$

… where P_{cr} is the buckling load in compression, P_x the applied compression load, and A_{ij} the terms of the laminate in-plane stiffness matrix.

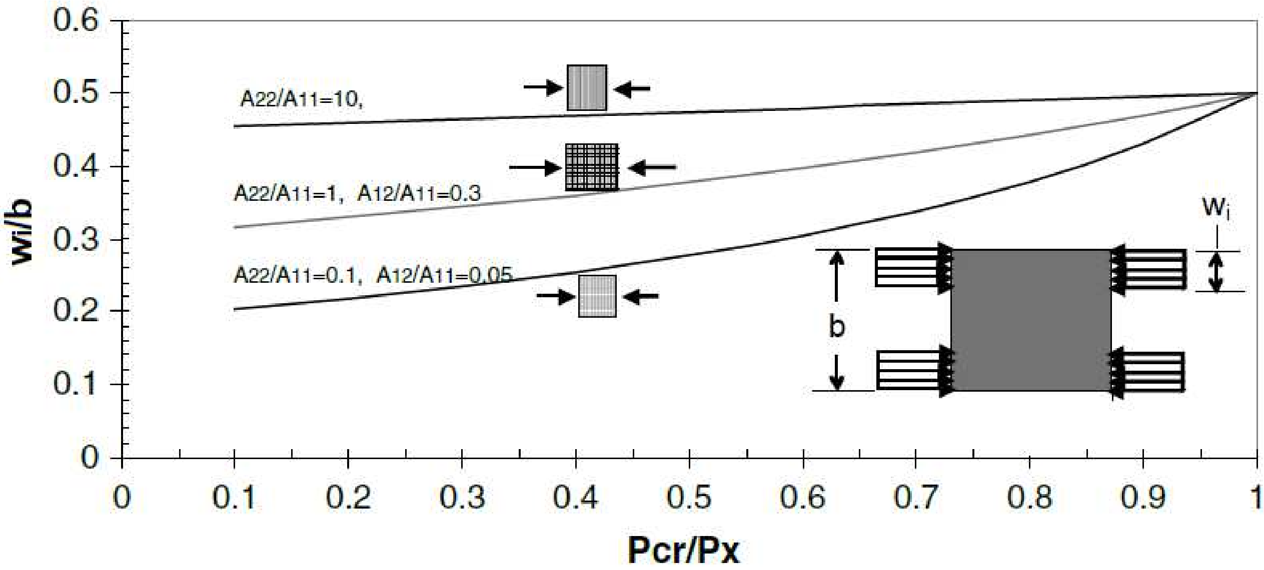

Figure 3. Variation of w_i as a function of loading fraction and material properties (Kassaploglou)

A typical value of w_i of the skin in compression, after post-buckling, will be around 0.3 times the width. If the skin is in tension or has not buckled, obviously the effective semi-width is 0.5 times the width.

The validity of abovementioned expression for skin panels with higher aspect ratios or some curvature (as in conventional fuselages), when the critical buckling load used in the formula was obtained for the specific configuration, is generally accepted as a reasonable approximation within the scope of applicability of non-numerical (non-high-fidelity) methodologies.

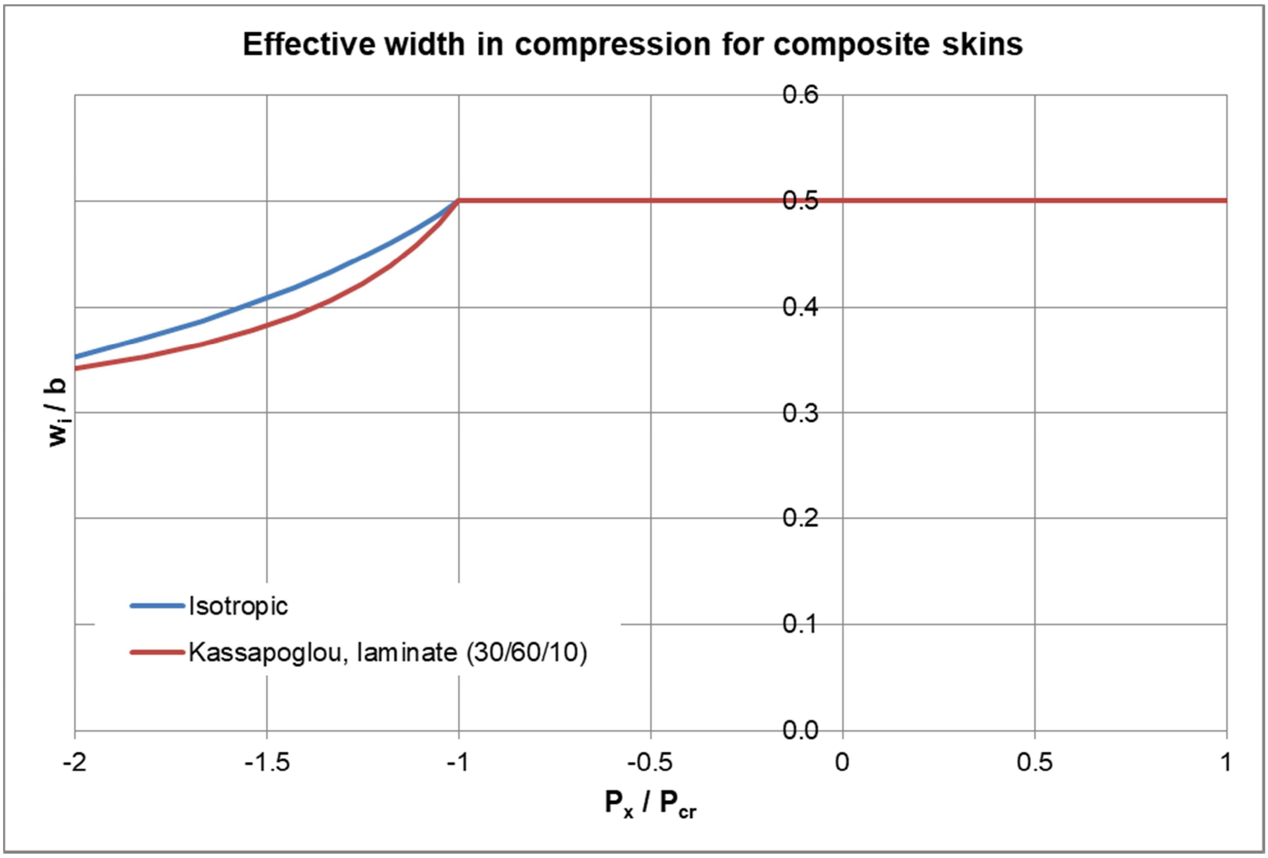

Figure 4. Differences in presented methods for effective skin (P_x > 0 means tension in this graph)

Note both approaches are relatively similar for typical design regions and traditional post-buckling policies (ultimate load exceeding the buckling onset but not above ratios around of P_x/P_{cr} ~ 1.5 in the most permissive cases).

<u>2.1.3 Special considerations for padded skins and bonded stringers</u>

It will normally be useful to remove spurious excesses of conservatism in calculation of the effective width of the skin in compression by taking into account the additional stiffness provision available in the joint between the stringer and the skin.

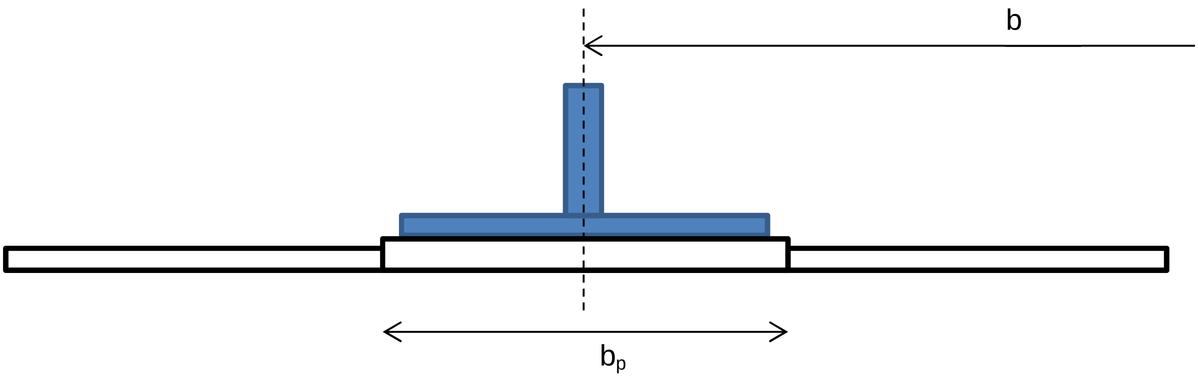

Figure 5. Padded skin and tee stringer

For tee, I or J stringer cross-sections (with their foot flanges centred in the stringer web), the semi-effective skin shall be calculated as:

$$w_i = max\ [\ min(\ b_p/2,\ w_p\ )\ ,\ w_{sk\text{-}i}\ ]$$

where:

w_i is the effective semi-width of skin (at each side of the stringer)

w_p is the effective semi-width calculated for the pad laminate (per §2.1.2)(*)

$w_{sk\text{-}i}$ is the effective semi-width calculated for the skin laminate (per §2.1.2)

(*) If the stringer foot is bonded/co-cured to the skin pad, the calculation of w_p shall be done with laminate resulting from joining pad and foot laminates.

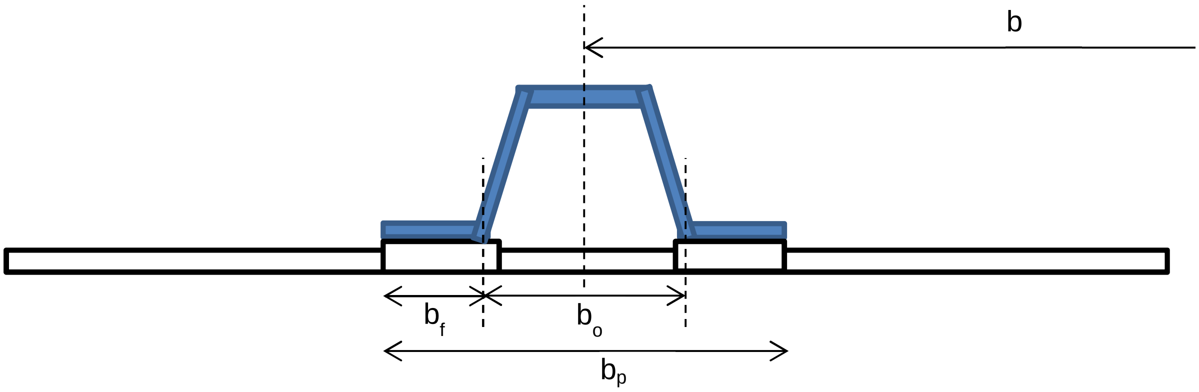

Figure 6. Padded skin and hat/omega stringer

For hat/omega cross-sections of width low compared with the panel width (then with the skin segment of the closed section very stable)(*), the semi-effective skin can be calculated as:

$$w_i = b_o/2 + \ max[\ min(\ b_f,\ w_p\ )\ ,\ min(\ w_{sk\text{-}i}\ ,\ (b\text{-}b_o)/2\ )\ ]$$

(*) In other words, the local buckling of the skin segment of width b_e , enclosed between the omega walls, is not allowed below the section/column failure.

Same abovementioned considerations regarding bonding and meaning of symbols apply here.

2.2 EFFECTIVE WIDTH OF SKIN ON CURVED FRAMES

Reserved for future use.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
[A]	N/mm	In-plane stiffness matrix of the laminate
b	mm	Skin width, distance between stiffeners
t	mm	Thickness of the skin
N_x	N/mm	Compression load (per unit width) in the skin (at a certain Y location)
P_{cr}	N	Buckling load force in compression of the whole cross section
P_x	N	Applied compression load force on the whole cross section
w_i	mm	Semi-effective skin width at one side of the stiffener
w	mm	Effective skin width
X	-	Longitudinal direction
Y	-	Transverse direction
ε_{cr}	-	Strain at buckling load in compression
ε_x	-	Applied compression strain

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	NT-T-ADS-97013	Ed. A	Manual Teórico del programa ARPA, versión 2.0
Ref. 2	[C. Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures
Ref. 3	[M. Schagerl]	2011	Analytical formulas for the effective width caused by bulging and flattening of curved aircraft panels

SOFTWARE

Program	***Version***	***Description***

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 04/05/2020	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEASP-TL3)	G. Baños 30/04/2020	Florian Glock 04/05/2020

Table 3. *Approval signatures table for last revision of this chapter*